

GOURAV SARKAR

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Skills

Languages: Java, JavaScript, Python, C++, C

Frameworks: Spring Boot, Node.js, React, Flutter, Flask, Express.js

Cloud & Tools: GCP, Azure, Docker, Git, Jenkins, CI/CD, MySQL, NoSQL, Firebase

AI/ML: AI Agents, LLMs, RAG, Distributed Systems, Vector Databases, Prompt Engineering

Experience

Software Engineer | [ServiceNow](#)

Hyderabad, India · 11/2023 – Present

- Resolved a heap memory bottleneck surfaced by **AT&T** (18,000+ assignment groups, **155M+ data points**), reducing historical data processing time by **61% (18 to 7 hours)** via time-series data model redesign and batch processing, fix deployed across **100+ Fortune 500 customers**.
- Improved forecast granularity **4x (hourly to 15-min intervals)** and reduced agent idle time by **30%** by introducing **TTR (Time to resolve) based staffing forecasting**, replacing coarse resource conversion formulas with granular headcount predictions.
- Redesigned time-series interval anchoring to fix a UTC-offset defect raised by DHL**, where interval boundaries misaligned under non-standard offsets (e.g. Nepal's +5:45); re-anchored the aggregation grid to local wall-clock time instead of offset-shifting UTC buckets, correcting **15/30/60-min forecasts** across **all global timezones**.
- Reduced manual shift planning overhead by **~50% for thousands of enterprise managers** by building **production LLM-powered forecast and shift planning AI agents** enabling natural language staffing operations.
- Led cross-team integration of **Workforce Management AI agents**, surfaced as a **unified conversational interface** consolidating **5+ workforce management operations** for **100+ Fortune 500 enterprise customers**.
- Delivered real-time operational visibility to **25,000+ enterprise teams** by revamping shift planning infrastructure including recurring events, timezone-aware holiday calendars, and **10+ manager dashboard widgets** (CSAT, SLA, P1 cases, agent utilization, skill demand) refreshing every **30 seconds**, reducing dashboard load time by **40%**.

Software Engineer | [Persistent Systems](#)

Pune, India · 02/2022 – 10/2023

- Built scalable **Java/Spring Boot APIs and MySQL** backend for **NORC at the University of Chicago**, integrating Excel-based survey data for US federal agencies (**CDC, NIH, VA**) with automated scheduling via Cron.
- Maintained production reliability through **CI/CD pipelines**, load/stress testing, and query optimization; ensured data integrity and security across federal agency deployments.

Projects

DevForge - Local-first CLI tool that keeps your entire codebase queryable via natural language, with no API keys or internet required. Scans and chunks repos into ~2000-char segments; hybrid BM25 + vector retrieval over incrementally indexed repos with SHA-256 delta detection, returning cited answers (file:line-range) via any Ollama-compatible model with zero cloud dependency. [GitHub](#)

GlimpseDesk - VS Code-style desktop media and file explorer built with Electron. Multi-tab, multi-window interface with native OS folder picker, video preview, and markdown rendering. Fully local, no cloud dependency. [GitHub](#)

Publications & Achievements

- Gourav Sarkar** et al. (2026). [CopperNet: Hybrid Deep Learning and LLM-Based Optimization for Copper Refining Processes](#). *ICAART 2026, Marbella, Spain*. [Published]
- Gourav Sarkar** et al. (2026). Agentic AI for Industrial Cooling Systems: Deep RL-Based Optimization with LLM-Enhanced Decision Support. *ISUW 2026, New Delhi*. [Accepted]
- Dhargawe, S., De Sarker, A., Jose, A., & **Sarkar, G.** (2024). [Multi-Severity Corrosion Classification Using Computer Vision](#). *ICSOT-INDIA 2023*. [Published]
- [All India Automobile Hackathon](#): Won 2nd prize out of 1,862 teams nationwide for a VR app. Organized by NEC, Mitsubishi Corporation and Isuzu Motors. (03/2022)

Education

Bachelor of Technology, CSE | [West Bengal University of Technology](#)

Kolkata, India · 07/2018 – 06/2022

- Major in Computer Science and Engineering **CGPA: 9.39/10**

Certifications

- [GCP Professional Cloud Architect](#) (01/2023)
- [GCP Associate Cloud Engineer](#) (10/2022)
- [Azure Fundamentals AZ-900](#) (06/2022)